

# EEE4114F: Problem Set 5 (DFT)

1. Compute the DFT of each of the following finite-length sequences considered to be of length  $N$  (where  $N$  is even):

(a)  $x[n] = \delta[n]$

(b)  $x[n] = \delta[n - n_0], 0 \leq n_0 \leq N - 1$

(c)  $x[n] = \begin{cases} 1, & n \text{ even}, & 0 \leq n \leq N - 1 \\ 0, & n \text{ odd}, & 0 \leq n \leq N - 1 \end{cases}$

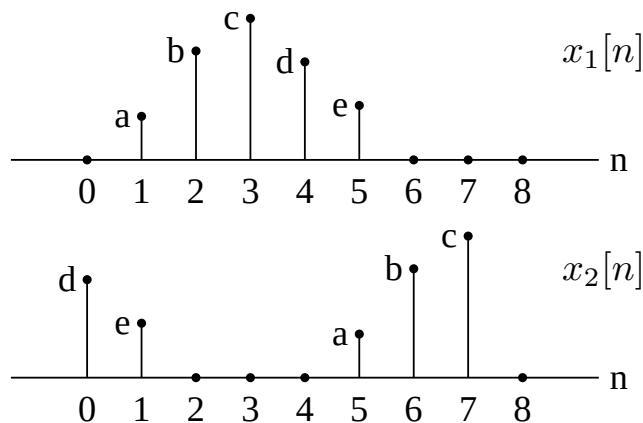
(d)  $x[n] = \begin{cases} 1, & 0 \leq n \leq N/2 - 1 \\ 0, & N/2 \leq n \leq N - 1 \end{cases}$

(e)  $x[n] = \begin{cases} a^n, & 0 \leq n \leq N - 1 \\ 0, & \text{otherwise.} \end{cases}$

2. Consider the complex sequence

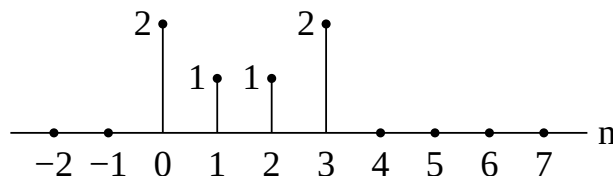
$$x[n] = \begin{cases} e^{j\omega_0 n}, & 0 \leq n \leq N - 1 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the Fourier transform  $X(e^{j\omega})$  of  $x[n]$ .
- (b) Find the  $N$ -point DFT  $X[k]$  of the finite-length sequence  $x[n]$ .
- (c) Find the DFT of  $x[n]$  for the case  $\omega_0 = 2\pi k_0/N$ , where  $k_0$  is an integer.
3. The two 8-point sequences  $x_1[n]$  and  $x_2[n]$  shown below have DFTs  $X_1[k]$  and  $X_2[k]$  respectively:



Determine the relationship between  $X_1[k]$  and  $X_2[k]$ .

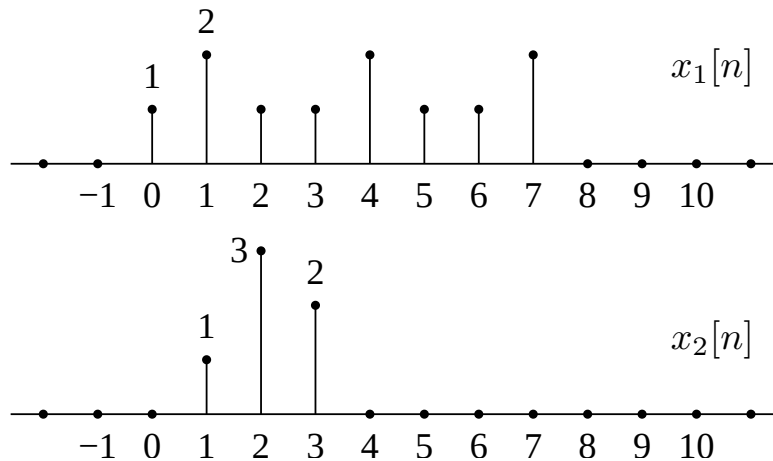
4. Consider the following finite-length sequence  $x[n]$ :



The five-point DFT of  $x[n]$  is denoted by  $X[k]$ . Plot the sequence  $y[n]$  whose DFT is

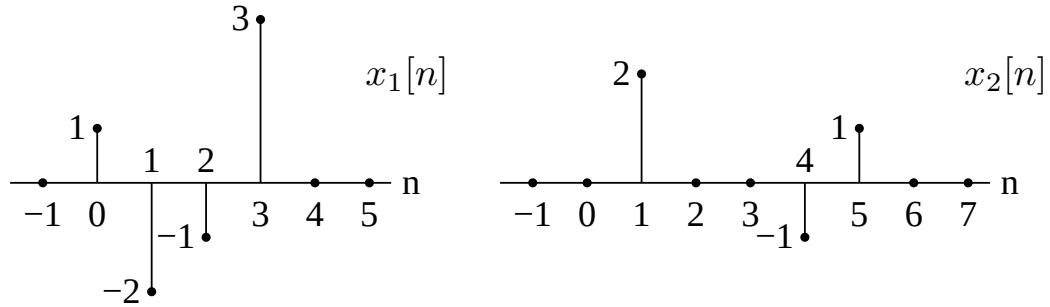
$$Y[k] = W_5^{-2k} X[k].$$

5. Two finite-length sequences are shown below:



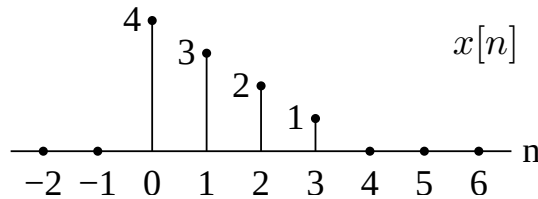
Assume that  $x_1[n]$  and  $x_2[n]$  are zero outside of the region shown in the figure. Let  $x_3[n]$  be the 8-point circular convolution of  $x_1[n]$  with  $x_2[n]$ . Determine  $x_3[2]$ .

6. The following figure shows two finite-length sequences  $x_1[n]$  and  $x_2[n]$ .



What is the smallest  $N$  such that the  $N$ -point circular convolution of  $x_1[n]$  and  $x_2[n]$  is equal to the linear convolution of these sequences?

7. Consider the real finite-length sequence



- (a) Sketch the finite-length sequence  $y[n]$  whose six-point DFT is

$$Y[k] = W_6^{4k} X[k],$$

where  $X[k]$  is the six-point DFT of  $x[n]$ .

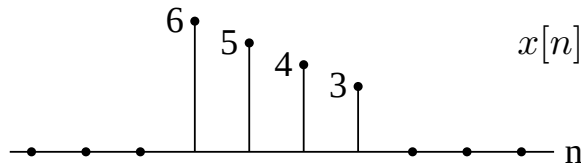
- (b) Sketch the finite-length sequence  $w[n]$  whose six-point DFT is

$$W[k] = \text{Re}\{X[k]\}.$$

- (c) Sketch the finite-length sequence  $q[n]$  whose three-point DFT is

$$Q[k] = X[2k], \quad k = 0, 1, 2.$$

8. Consider the following finite-length sequence:



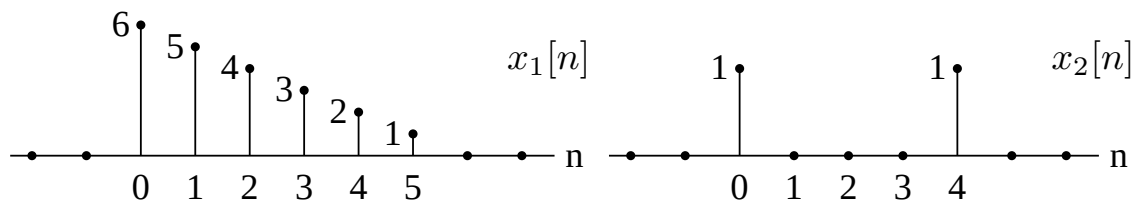
Sketch the sequences

$$x_1[n] = x[((n-2))_4], \quad 0 \leq n \leq 3,$$

and

$$x_2[n] = x[((-n))_4], \quad 0 \leq n \leq 3.$$

9. Sketch the  $N$ -point circular convolution of the following two sequences



for  $N = 6$  and  $N = 10$ .